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Chapter nine

MS Business Intelligence

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1. Intro to Business Intelligence

- ال **Business Intelligence** هي أدوات يتم تزويدنا بها لتحليل البيانات لإستخراج معلومات و بموجبها نحصل على معرفة
- يتم الإستفادة من المعلومات و المعرفة لإتخاذ القرارات الإستراتيجية

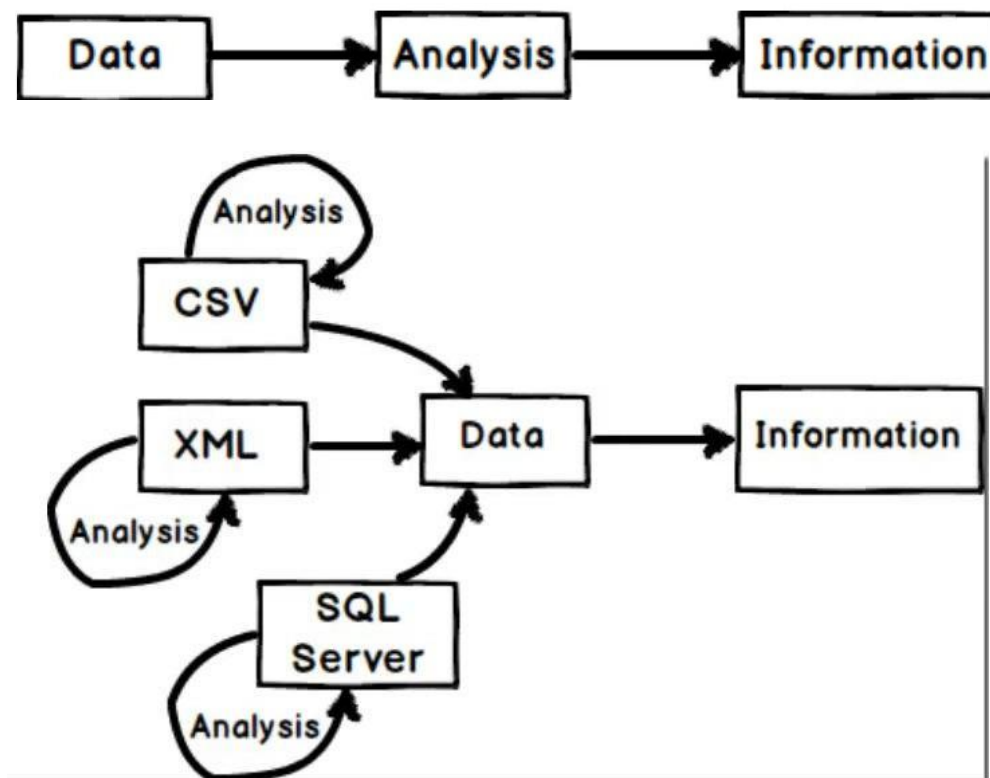
Data is raw, unorganized facts that need to be processed, numbers, texts, images or symbols that represent events or measures. Information is processed data.

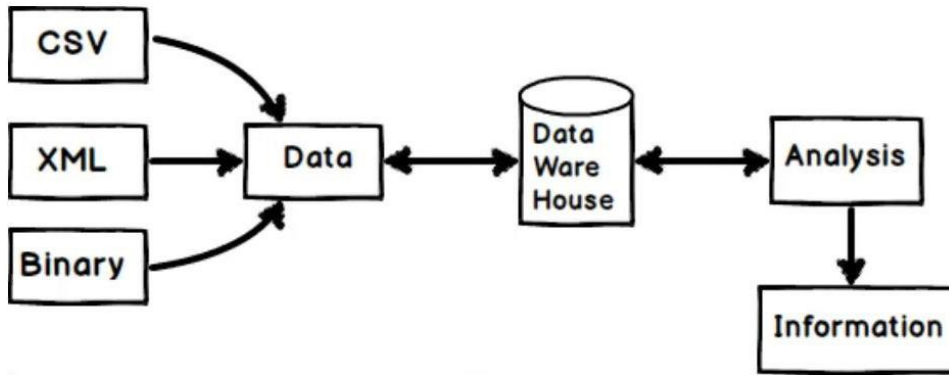
Business Intelligence (BI) refers to technologies, applications and practices for the collection, integration, analysis, and presentation of business information. The purpose of Business Intelligence is to support better business decision making.

If a picture is worth a thousand words, the following illustration by pictures might be sufficient and worth a lot of papers

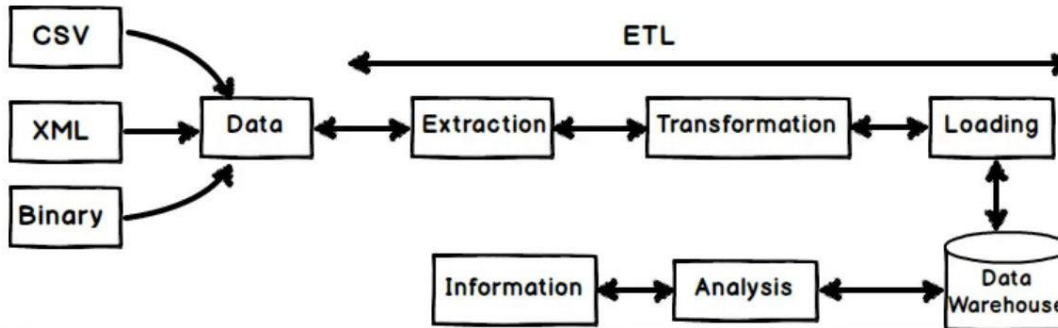
- يتم تحويل ال Data الخام عبر تحليلها إلى Information معلومات

- هناك مستوى أعلى من المعرفة Knowledge التي نحصل عليها من خلال المعلومات Information و هو الحكمة Wisdom

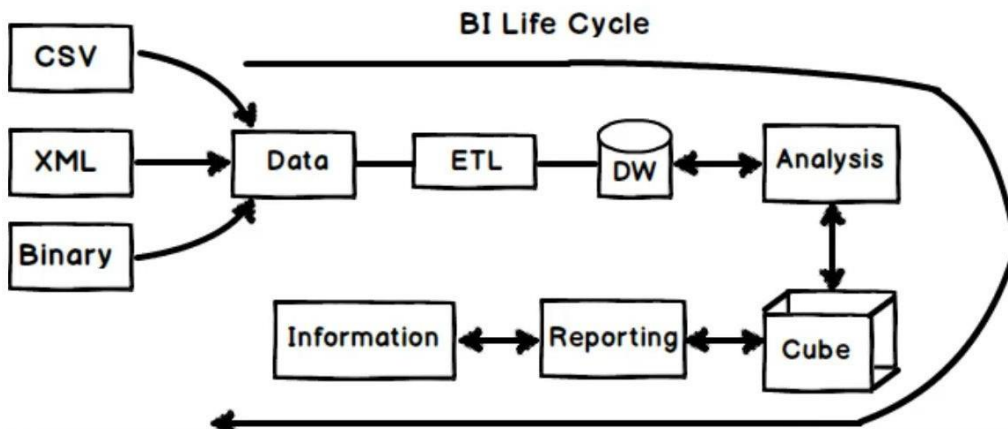




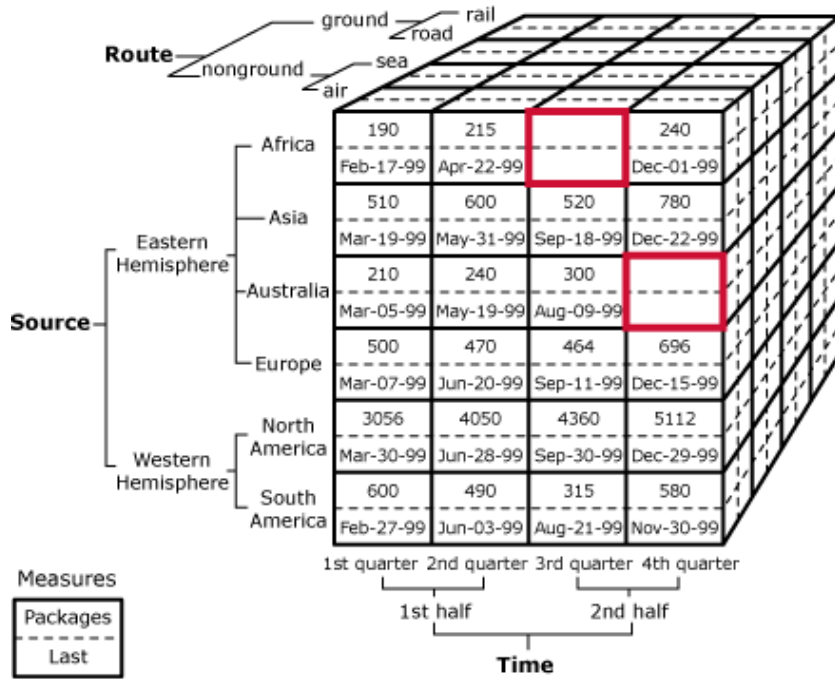
- يطلق على عملية نقل ال Data بأشكالها و تنسيقاتها المختلفة إلى Datawarehouse بعملية ETL – Extraction Transformation Loading حيث يتم بها إستخراج ال Data من بيئتها السابقة و تحويلها لنمط مشترك مع كل ال Data المستخرجة الأخرى و من ثم تحميلها إلى ال Datawarehouse



- دورة حياة ال Business Intelligence:



حيث أن ال Cube هو عبارة عن عملية Denormalization لل Data

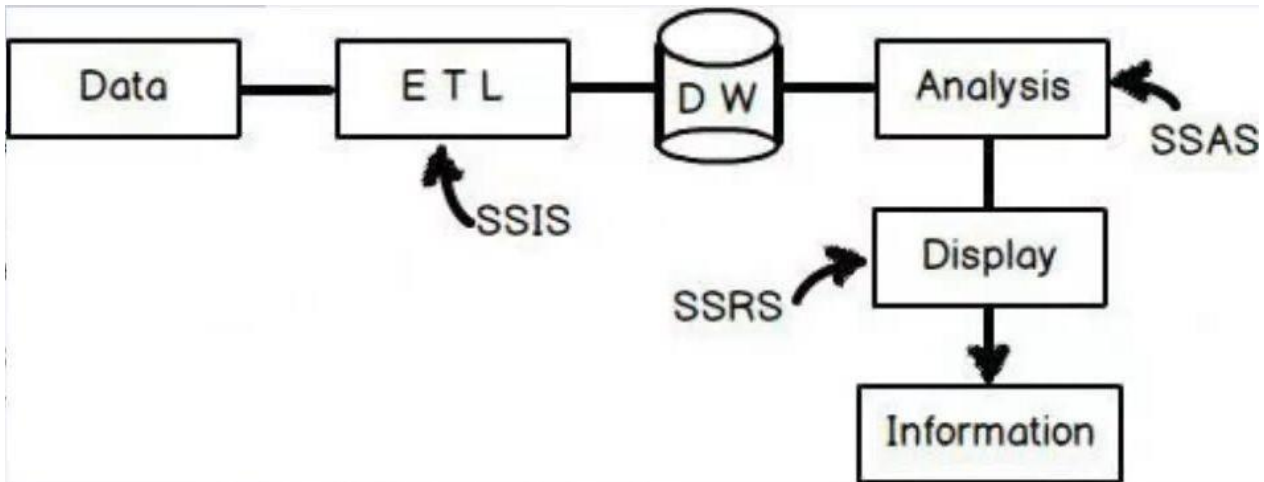


- في ال SQL Server لدينا 3 أدوات تساعدنا في العمل على ال Business Intelligence و هي:

SSIS – SQL Server Integration Service: المسؤولة عن عمليات ال ETL

SSAS – SQL Server Analysis Service: المسؤولة عن التحليل

SSRS – SQL Server Reporting Service: التي تقوم بعرض نتائج التحليل (ليست إجبارية حيث لدينا أدوات أخرى تقوم بهذه الوظيفة)



2. Server Analysis Services SSAS

- لها شكلين:

(1) Multidimensional Model (الأقدم و الافتراضي و الذي يحوي خدمات أكثر و الذي يستهلك موارد أكثر)
(2) Tabular Model

Analytical data engine used in decision support and business analytics. It provides enterprise-grade semantic data models for business reports and client applications such as Power BI, Excel, Reporting Services reports, and other data visualization tools. A typical workflow includes creating a tabular or multidimensional data model project in Visual Studio, deploying the model as a database to a server instance, setting up recurring data processing, and assigning permissions to allow data access by end-users. When it's ready to go, your semantic data model can be accessed by client applications supporting Analysis Services as a data source.

- توجد خدمات ال SSAS في كل من SQL Server بنوعيه (Azure, On-premises)

- ❖ **Azure Analysis Services** - Supports tabular models at the 1200 and higher compatibility levels. DirectQuery, partitions, row-level security, bi-directional relationships, and translations are all supported.
- ❖ **SQL Server Analysis Services** - Supports tabular models at all compatibility levels, multidimensional models, data mining, and Power Pivot for SharePoint.

2.1. Tabular models

- تأخر إصداره بشكل رسمي عن ال **Multidimensional Model** بسبب إستهلاكه الكبير للذاكرة حيث يقوم وضع ال **Data** في الذاكرة, ثم يضغطها 10 مرات ليعطي أداء أفضل ب 10 مرات
 - يعتمد ال **Tabular Model** في عمله على ال **Columned-store-index** الذي يعمل على الذاكرة
 - ينحصر بإستخدام ال **Tabular Model** عند وجود ذاكرة كبيرة و سريعة فقط
 - يستخدم إحدى تقنيتين:

(1) In-Memory: يقوم بوضع كل ال **Data** في ال **Memory**

(2) Direct-Query: تستخدم عند عدم وجود ذاكرة كافية, حيث يستخدم بها طريقة ال **Multithreading** للقراءة بسرعة كبيرة بدون نسخ البيانات إلى الذاكرة

- سابقا كان يطلق على طريقة ال **In-Memory** اسم **Vertipaq**, و حاليا تم تعديل الاسم إلى **Velocity x** التي توفر وصول سريع للعناصر و البيانات في ال **Tabular Model**

- في ال **Tabular Model** نستطيع إستخدام لغة إسمها **Dax** (من نوع **Formula Language**) التي توفر **Row Level Security** و توفر **Measuring** (عمليات حسابية على ال **Data**)
 - يتم إنشاء ال **Tabular Model** بإستخدام ال **SQL Server Data Tools – SSDT** في ال **SQL Server**

Tabular models in Analysis Services are databases that run in-memory or in DirectQuery mode, connecting to data directly from back-end relational data sources. By using state-of-the-art compression

algorithms and multi-threaded query processor, the Analysis Services Vertipaq analytics engine delivers fast access to tabular model objects and data by reporting client applications like Power BI and Excel.

While in-memory models are the default, DirectQuery is an alternative query mode for models that are either too large to fit in memory, or when data volatility precludes a reasonable processing strategy.

DirectQuery achieves parity with in-memory models through support for a wide array of data sources, ability to handle calculated tables and columns in a DirectQuery model, row level security via DAX expressions that reach the back-end database, and query optimizations that result in faster throughput.

DAX is a formula language. You can use DAX to define custom calculations for Calculated Columns and for Measures (also known as calculated fields).

Tabular models are created in SQL Server Data Tools (SSDT) using the Tabular model project template. The project template provides a design surface for creating semantic model objects like tables, partitions, relationships, hierarchies, measures, and KPIs.

2.2. Multidimensional models

- هو الأقدم و الافتراضي, يسمح بالتعامل مع بيانات باستخدام:

MOLAP نقوم عن طريقه بإنشاء Cube لل Data تحوي مجاميع ال Data بكل أشكالها جاهزة لأي Query

ROLAP, الذي يحتوي على Cube فارغ, مرتبط بال Relational Database (العلائقية) و يقوم بتعبئة ال Cube الخاص به عند الحاجة

HOLAP التي هي عبارة عن OLAB من نوع Hybrid ,

OLAP (Online Analysis Processing) الذي يسمح بتحليل مباشر على ال Data التي سنقوم ب Cube لها بواسطة ال (MOLAR

- توفر أمان من نوع Cell-Level-Security

An Analysis Services multidimensional solution uses cube structures for analyzing business data across multiple dimensions. Multidimensional mode is the default server mode of Analysis Services. It includes a query and calculation engine for OLAP data, with MOLAP, ROLAP, and HOLAP storage modes to balance performance with scalable data requirements.

2.2.1. Benefits of Multidimensional Solutions

- من أهم مميزات ال Multidimensional Model:

توفير أداء قوي بدون الحاجة لموارد كثيرة, و تعتبر أفضل في التعامل مع البيانات و الإستعلامات المعقدة

توفر مصدر واحد للبيانات مما يسمح بأداء أعلى و تحليل أسرع للبيانات

توافقها مع كل أدوات التقارير الخاصة بال BI مثل Excel, Reporting Services, PerformancePoints

The primary reason for building an Analysis Services multidimensional model is to achieve fast query performance against business data. A multidimensional model is composed of cubes and dimensions that can be annotated and extended to support complex query constructions.

BI developers create cubes to support fast response times, and to provide a single data source for business reporting. Given the growing importance of business intelligence across all levels of an organization, having a single source of analytical data ensures that discrepancies are kept to a minimum, if not eliminated entirely.

Another important benefit to using Analysis Services multidimensional databases is integration with commonly used BI reporting tools such as Excel, Reporting Services, and PerformancePoint, as well as custom applications and third-party solutions.

2.2.2. Multidimensional Model Databases (SSAS)

An Analysis Services database is a collection of data sources, data source views, cubes, dimensions, and roles. Optionally, an Analysis Services database can include structures for data mining, and custom assemblies that provide a way for you to add user-defined functions to the database.

Cubes are the fundamental query objects in Analysis Services. When you connect to an Analysis Services database via a client application, you connect to a cube within that database. A database might contain multiple cubes if you are reusing dimensions, assemblies, roles, or mining structures across multiple contexts. You can create and modify an Analysis Services database programmatically or by using one of these interactive methods:

- ❖ Deploy an Analysis Services project from SQL Server Data Tools (SSDT) to a designated instance of Analysis Services. This process creates an Analysis Services database, if a database with that name does not already exist within that instance, and

instantiates the designed objects within the newly created database. When working with an Analysis Services database in SQL Server Data Tools, changes made to objects in the Analysis Services project take effect only when the project is deployed to an Analysis Services instance.

- ❖ Create an empty Analysis Services database within an instance of Analysis Services, by using either SQL Server Management Studio or SQL Server Data Tools (SSDT), and then connect directly to that database using SQL Server Data Tools (SSDT) and create objects within it (rather than within a project). When working with an Analysis Services database in this manner, changes made to objects take effect in the database to which you are connecting when the changed object is saved.

SQL Server Data Tools (SSDT) uses integration with source control software to support multiple developers working with different objects within an Analysis Services project at the same time. A developer can also interact with an Analysis Services database directly, rather than through an Analysis Services project, but the risk of this is that the objects in an Analysis Services database can become out of sync with the Analysis Services project that was used for its deployment. After deployment, you administer an Analysis Services database by using SQL Server Management Studio. Certain changes can also be made to an Analysis Services database by using SQL Server Management Studio, such as to partitions and roles, which can also cause the objects in an Analysis Services database to become out of sync with the Analysis Services project that was used for its deployment.

من أهم الفروقات بين ال Tabular, Multidimensional models : جميع الجداول هامة للحفاظ

Feature	Multidimensional Modeling	Tabular Modeling
Data Sources	Relational Databases	Relational databases, Analysis Services cubes, Reporting Services reports, Azure DataMarket datasets, data feed, Excel files and text files
Development Tool	SQL Server Data Tools (SSDT)	SQL Server Data Tools (SSDT)
Management Tool	SQL Server Management Studio (SSMS)	SQL Server Management Studio (SSMS)
API	AMO, AMOMD.NET and PowerShell	AMO, AMOMD.NET and PowerShell
Reporting & Analysis Tool	Report Builder, Report Designer, Excel PivotTable and PerformancePoint dashboard	Report Builder, Report Designer, Excel PivotTable, PerformancePoint dashboard and Power View
Query & Expression Language	MDX Scripting – Calculations (cube designer) MDX – Queries & Calculations DMX – Data Mining Queries	DAX – Calculations & Queries MDX – Queries
Write Back Support	Yes	No
Security	Role-based permissions in SSAS & cell-level security	Role-based permissions in SSAS and row-level security
Data Compression	3+ times	10+ times

Scalability

Multidimensional

- Pre-Aggregated Data From Disk
- Can Store Very Large Amounts of Data
- Uses Aggregations to Increase Query Performance
- Data Compression 3x
- MOLAP - Balanced configuration – Disk I/O & RAM
- ROLAP – Maximize Disk I/O, Minimize network latency

Tabular

- In-Memory Technology (x-Velocity)
- Can Store Large Amounts of Data
- No Aggregations. Column-Based Storage.
- Data Compression 10x
- In Memory - Emphasize RAM & CPUs with fast clock speeds.
- Direct Query – Relational DB Performance, Low Network Latency, Faster CPUs

Summary

Consider Tabular...

- If You Have a Short Development Timeline
- If You are Working with a Plethora of Memory
- If Your Data Model is Simple
- If You Have Many Disparate Data Sources
- If Users Need to Query Large Amounts of Detail Data

Consider Multidimensional...

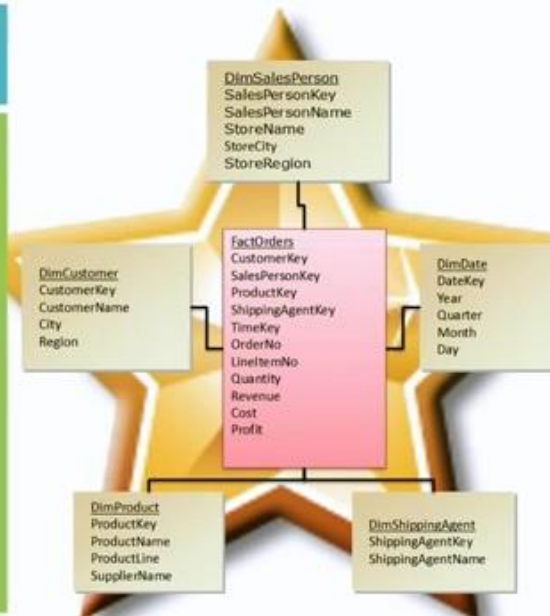
- If You are Using SQL Server 2008 R2 or Earlier
- If You Have a Multi-Terabyte Data Source
- If You Have a Complex Data Model
- If You Need Multidimensional only features (Actions, Data Mining, Writeback, etc.)

2.3. Data Warehouse Schemas

- في ال Star Schema: جدول الحقيقة يتصل مع Dimensions بشكل إتصال نجمة يحوي كل منها على Table واحد

Star Schema

- A star schema has a single table for each dimension
- Each table supports all attributes for that dimension
- Typically a de-normalized solution



- في ال Snowflake Schema: يستطيع كل Dimension متصل مع جدول الحقيقة, أن يتصل مع جداول أخرى

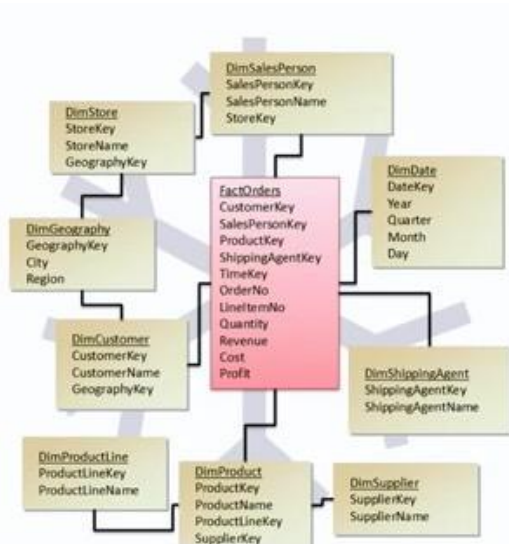
و ينخفض مستوى الأداء فيها كلما كثر عدد الأفرع المتصلة مع ال Dimensions, حيث للإستعلام عن كل فرع متصل مع أحد

ال Dimensions يجب استخدام ال Join مع جميع الأفرع التي قبله للوصول إلى جدول الحقيقة

- نستطيع القيام ب Normalization لجميع الأفرع المتصلة عن طريق Dimension واحد لدمجها إلى DB واحدة

Snowflake Schema

- More normalized solution
- Typically contains multiple tables per dimension
- Each table contains dimension key, value, and the foreign key value for the parent



- مثال عن بيانات جدول الحقيقة و ال *Dimensions* المتصلة به:

The screenshot displays the Microsoft Visual Studio interface with the 'Model.bim' window open. The window shows a data table with 12 rows and 9 columns. The columns are: ProductKey, OrderDate, DueDate, ShipDate, CustomerKey, PromotionKey, CurrencyKey, SalesOrderNumber, and SalesOrderID. The data is as follows:

	ProductKey	OrderDate	DueDate	ShipDate	CustomerKey	PromotionKey	CurrencyKey	SalesOrderNumber	SalesOrderID
1	528	20130128	20130209	20130204	14870	1	100	4	SO51900
2	528	20130129	20130210	20130205	15319	1	100	4	SO51948
3	528	20130131	20130212	20130207	16384	1	100	4	SO52043
4	528	20130131	20130212	20130207	15476	1	100	4	SO52045
5	528	20130201	20130213	20130208	15861	1	100	4	SO52094
6	528	20130203	20130215	20130210	26017	1	100	4	SO52175
7	528	20130203	20130215	20130210	14761	1	100	4	SO52190
8	528	20130204	20130216	20130211	22038	1	100	4	SO52232
9	528	20130204	20130216	20130211	22163	1	100	4	SO52234
10	528	20130204	20130216	20130211	16018	1	100	4	SO52245
11	528	20130205	20130217	20130212	25839	1	100	4	SO52301
12	528	20130205	20130217	20130212	11260	1	100	4	SO52314

The bottom status bar indicates 'Record: 1 of 60,398' and 'FactInternetSales'.

2.4. What is Data mining?

هي عملية التنقيب في البيانات الكبيرة للحصول على

الأنماط المخفية (Hidden patterns)

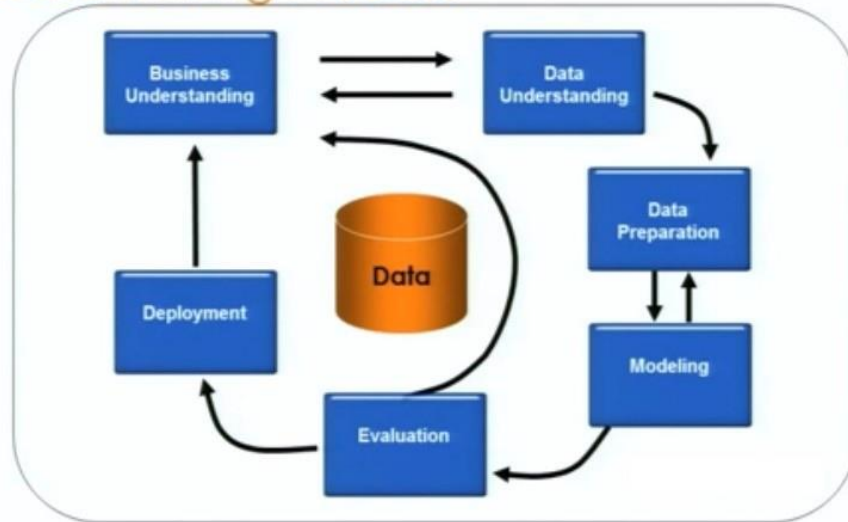
The practice of examining large databases in order to generate new knowledge. It's the process of analyzing hidden patterns.

It results with knowledge (summary of information which is a summary of data).

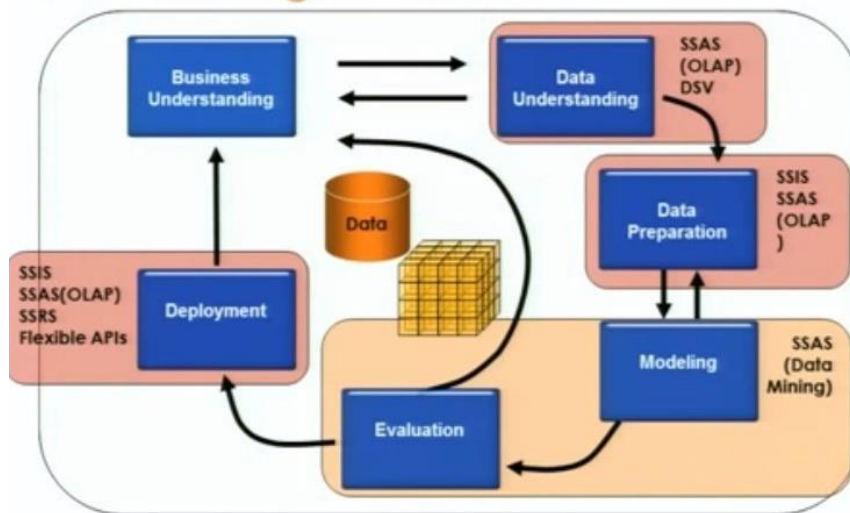
Data mining is:

- ❖ Exploring Data.
- ❖ Finding out Patterns.
- ❖ Performing Predictions.

Data Mining Process



Data Mining Process in SQL



2.4.1. Data Mining Algorithms

An *algorithm* in data mining (or machine learning) is a set of heuristics and calculations that creates a model from data. To create a model, the algorithm first analyzes the data you provide, looking for specific types of patterns or trends. The algorithm uses the results of this analysis over many iterations to

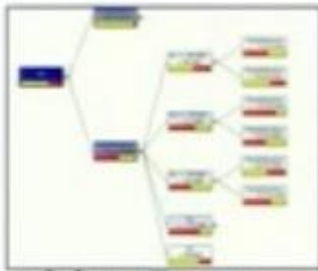
find the optimal parameters for creating the mining model. These parameters are then applied across the entire data set to extract actionable patterns and detailed statistics.

The mining model that an algorithm creates from your data can take various forms, including:

- A set of clusters that describe how the cases in a dataset are related.
- A decision tree that predicts an outcome, and describes how different criteria affect that outcome.
- A mathematical model that forecasts sales.
- A set of rules that describe how products are grouped together in a transaction, and the probabilities that products are purchased together.

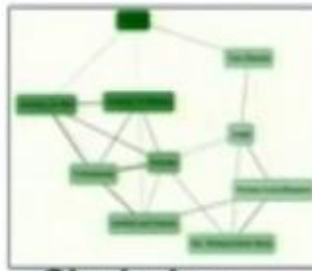
2.4.2. **Choosing an Algorithm by Type**

- **Classification algorithms** predict one or more discrete variables, based on the other attributes in the dataset.
- **Regression algorithms** predict one or more continuous numeric variables, such as profit or loss, based on other attributes in the dataset.
- **Segmentation algorithms** divide data into groups, or clusters, of items that have similar properties.
- **Association algorithms** find correlations between different attributes in a dataset. The most common application of this kind of algorithm is for creating association rules, which can be used in a market basket analysis.
- **Sequence analysis algorithms** summarize frequent sequences or episodes in data, such as a series of clicks in a web site, or a series of log events preceding machine maintenance.

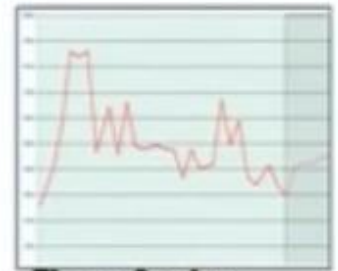


Decision Trees

Introduced in SQL Server 2000



Clustering



Time Series



Sequence Clustering



Association Rules

Class	Value	Class	Value
Class 1	0.0	Class 2	0.0
Class 1	0.1	Class 2	0.1
Class 1	0.2	Class 2	0.2
Class 1	0.3	Class 2	0.3
Class 1	0.4	Class 2	0.4
Class 1	0.5	Class 2	0.5
Class 1	0.6	Class 2	0.6
Class 1	0.7	Class 2	0.7
Class 1	0.8	Class 2	0.8
Class 1	0.9	Class 2	0.9
Class 1	1.0	Class 2	1.0

Naïve Bayes



Neural Networks

**+ Linear Regression
Logistic Regression**

2.4.3. Choosing an Algorithm by Task

To help you select an algorithm for use with a specific task, the following table provides suggestions for the types of tasks for which each algorithm is traditionally used.

Examples of tasks	Microsoft algorithms to use
Predicting a discrete attribute:	Microsoft Decision Trees Algorithm
Flag the customers in a prospective buyers list as good or poor prospects.	Microsoft Naive Bayes Algorithm
Calculate the probability that a server will fail within the next 6 months.	Microsoft Clustering Algorithm
Categorize patient outcomes and explore related factors.	Microsoft Neural Network Algorithm
Predicting a continuous attribute:	Microsoft Decision Trees Algorithm
Forecast next year's sales.	Microsoft Time Series Algorithm
Predict site visitors given past historical and seasonal trends.	Microsoft Linear Regression Algorithm
Generate a risk score given demographics.	
Predicting a sequence:	Microsoft Sequence Clustering Algorithm
Perform clickstream analysis of a company's Web site.	
Analyze the factors leading to server failure.	
Capture and analyze sequences of activities during outpatient visits, to formulate best practices around common activities.	
Finding groups of common items in transactions:	Microsoft Association Algorithm
Use market basket analysis to determine product placement.	Microsoft Decision Trees Algorithm
Suggest additional products to a customer for purchase.	
Analyze survey data from visitors to an event, to find which activities or booths were correlated, to plan future activities.	

3. SQL Server Integration Services SSIS

- هي التي تقوم بإنشاء **Packages** خاصة لنقل البيانات و القيام بعمليات ال **ETL**

Microsoft Integration Services is a platform for building enterprise-level data integration and data transformations solutions. Use Integration Services to solve complex business problems by copying or downloading files, loading data warehouses, cleansing and mining data, and managing SQL Server objects and data.

Integration Services can extract and transform data from a wide variety of sources such as XML data files, flat files, and relational data sources, and then load the data into one or more destinations.

Integration Services includes a rich set of built-in tasks and transformations, graphical tools for building packages, and the Integration Services Catalog database, where you store, run, and manage packages.

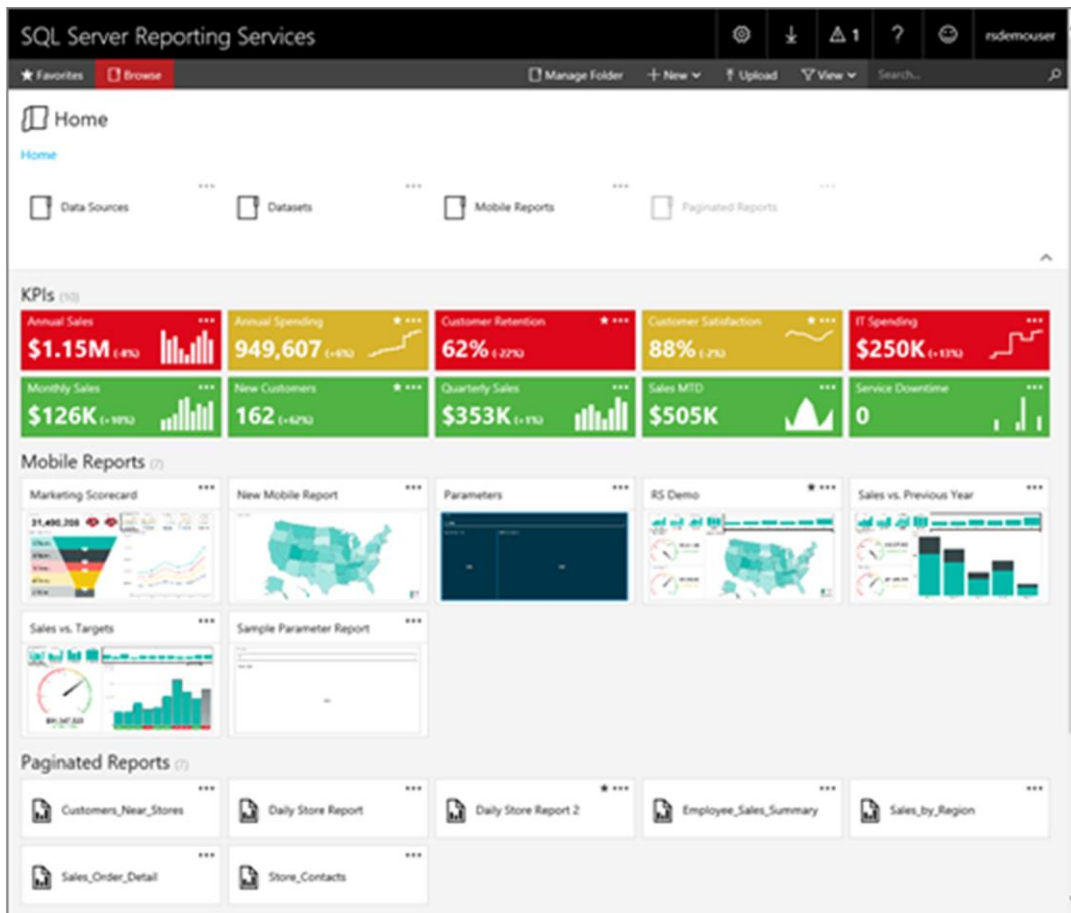
You can use the graphical Integration Services tools to create solutions without writing a single line of code. You can also program the extensive Integration Services object model to create packages programmatically and code custom tasks and other package objects.

4. SQL Server Reporting Service SSRS

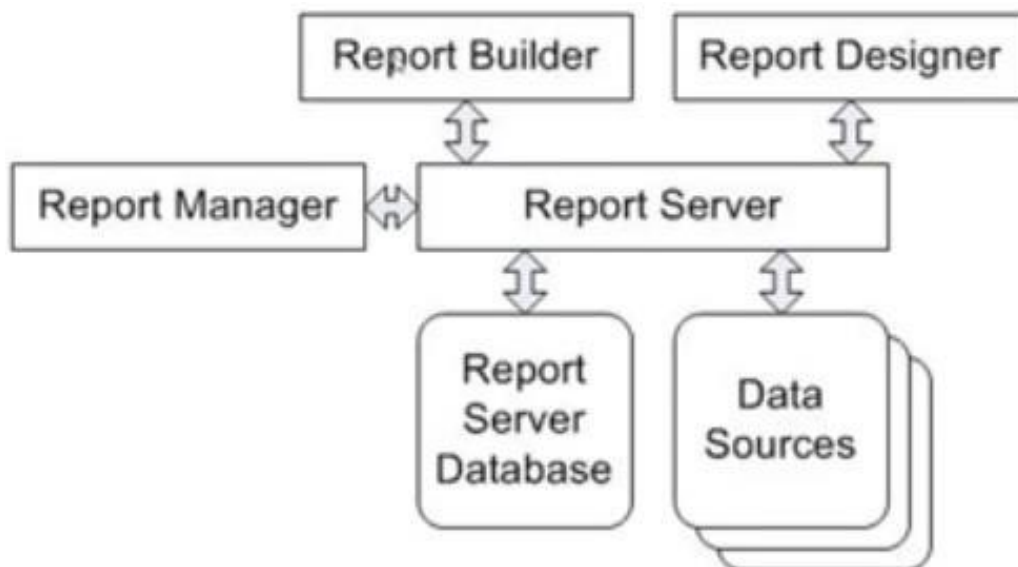
- هي جواب ال **SQL Server** عن ال **Report** كيف سيكون

SSRS is Microsoft's answer to business reporting. It provides a unified, server-based, extensible, and scalable platform from which to deliver and present information. Its scope extends from traditional paper reports to web-based delivery and interactive content. SSRS can also be configured to deliver reports to peoples' inboxes, file shares, and so on.

SSRS is capable of generating reports in various formats, such as the web-oriented Hypertext Markup Language (HTML) and desktop application (Microsoft Excel and CSV) formats, thus allowing users to manipulate their data in whatever format is required. In addition, SharePoint can be used as a front end for SSRS, allowing reports to be presented directly in corporate portals.



4.1. SSRS Architecture



4.1.1. Report Viewer

هي التي تحوي ال metadata و تعد مثل ال master DB في ال SQL Server

The primary mechanism for viewing reports over the Web

4.1.2. Report Builder

هو الذي نبني خلاله ال Report

The tool that provides users with a front end for ad hoc reporting against a SQL Server or Analysis Services database. Unlike most ad hoc reporting tools, users of Report Builder do not need to know Structured Query Language (SQL) or anything about joins or grouping to create reports.

4.1.3. Report Designer

هو الذي نبني خلاله ال Report

The tool that takes on the job of building advanced reports. Although Report Builder does a good job as an ad hoc reporting tool, Report Designer was made to tackle really advanced reports.

4.1.4. Report Server

هو عبارة عن Stateless Server

A report server is a stateless server that uses the SQL Server Database Engine to store metadata and object definitions. A native mode Reporting Services installation uses two databases to separate persistent data storage from temporary storage requirements, you can access it by navigating to the url: <http://SERVERNAME/reportServer>

4.1.5. Report Manager

هو عبارة عن Web-based Server يحوي جميع أنواع ال Reports الأخرى

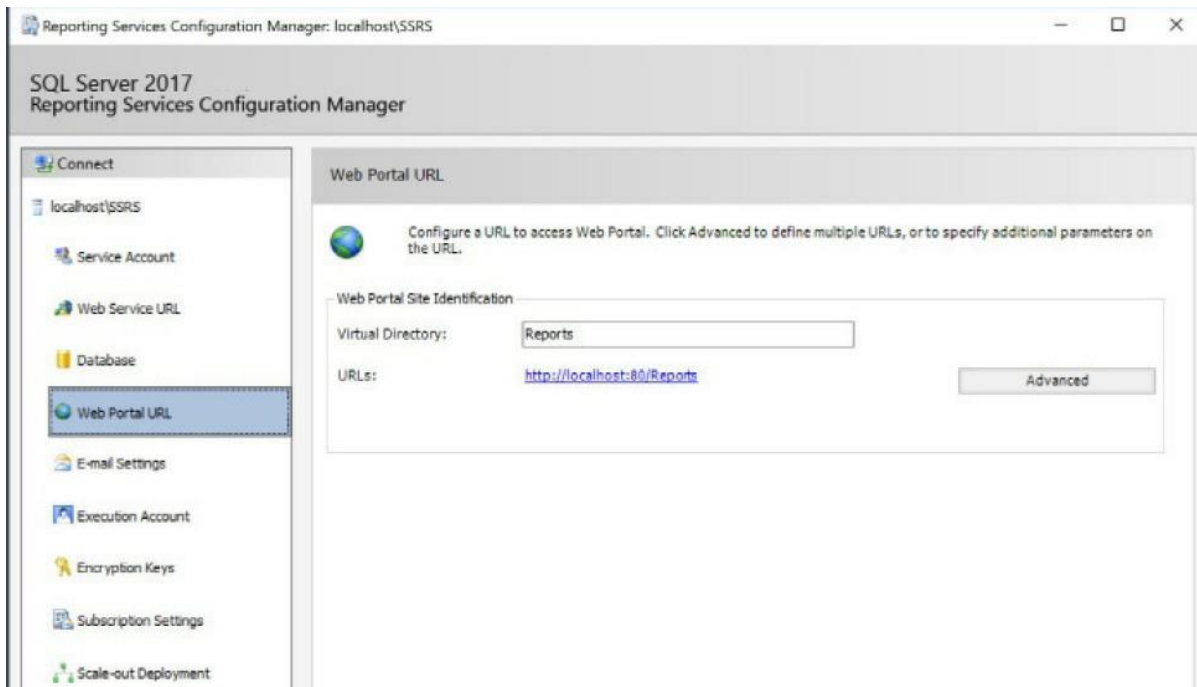
A Web-based report access and management tool that you use to administer a single report server instance from a remote location over an HTTP connection. You can also use Report Manager for its report viewer and navigation features, you can access it by navigating to the url: <http://SERVERNAME/reports>,

4.1.6. Report Services Configuration Manager

The Report Services Configuration Manager simplifies customization of the behavior of features and capabilities offered by SQL Server Reporting Services. You can use it to perform the following tasks and more:

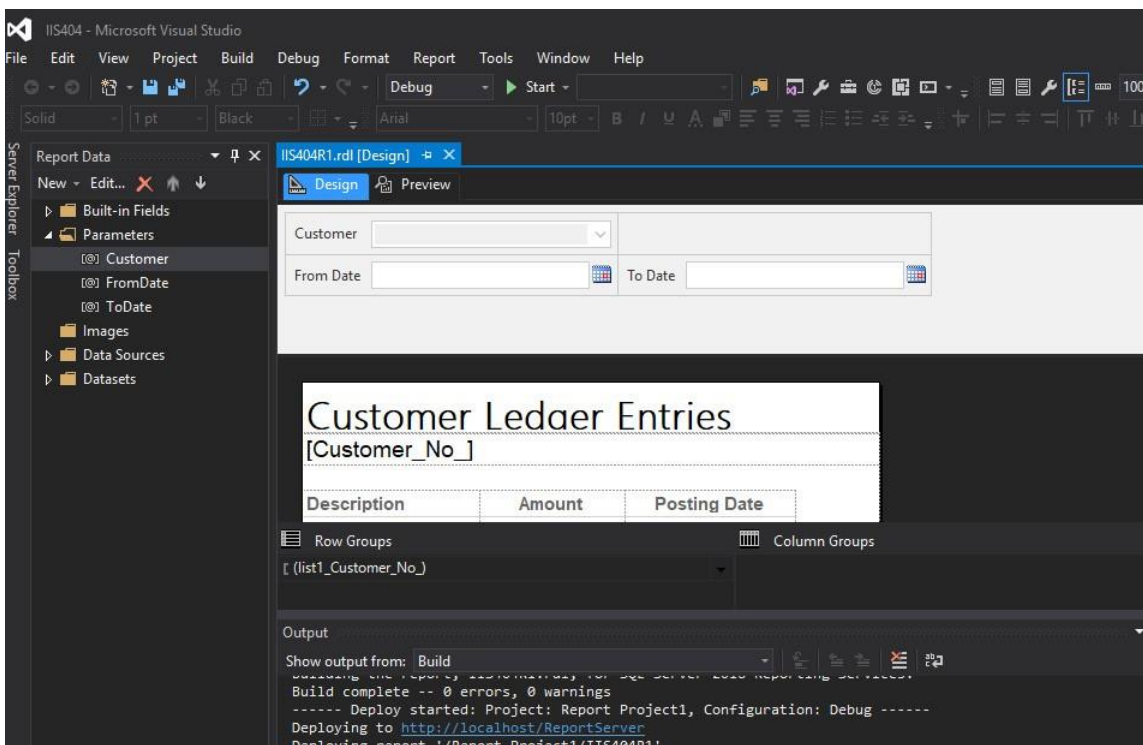
- Create or select existing Report Server databases
- Define the URLs used to access the Report Server and Report Manager

- Configure the Report Server Service Account
- Modify the connection string used by the Report Server
- Set up email distribution capability



4.2. Design

The designer is SQL Server data tools SSDT which is the development environment for all Microsoft Business intelligence members (SSIS, SSRS, and SSAS).



References

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